Does Syntactic Structure Make Language Models More Robust to Noise? LSTM vs. RNNG Subject–Verb Agreement Under Training-Data Corruption

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Abstract

1 Motivation

The core motivation of this study is to use controlled neural network experiments to investigate a long-standing debate in linguistics and cognitive science: how grammatical knowledge can be acquired from imperfect linguistic input.

According to Chomsky's poverty-of-the-stimulus argument, children are exposed to noisy and incomplete language containing speech errors, disfluencies, and grammatical violations, yet still acquire complex hierarchical grammatical systems. This observation has often been taken as evidence that language acquisition depends on innate linguistic structure rather than solely on statistical learning from input. However, because it is neither ethically nor practically possible to systematically manipulate the linguistic environments of human children, the extent to which grammar acquisition depends on innate structure remains difficult to evaluate through direct experimentation.

Recent advances in computational linguistics provide a new opportunity to address this problem. Neural language models offer a uniquely controllable experimental platform in which both the learner and the learning environment can be manipulated. The inductive biases of different architectures can be interpreted as different forms of internal structure, while the quality of linguistic input can be controlled through the construction of training corpora. By systematically introducing grammatical corruption into the training data, we can simulate increasingly degraded linguistic environments and observe how different learners respond.

Beyond the broader question of whether grammatical knowledge can be acquired from noisy input, this study also seeks to understand how noise affects the internal generalization strategies of language models. Natural language is fundamentally hierarchical, yet standard sequence models lack an explicit inductive bias for hierarchical structure. As grammatical evidence becomes increasingly unreliable, models may continue to rely on hierarchical syntactic representations, or they may gradually abandon them in favor of simpler linear heuristics based on surface word order. Tracking this transition provides a window into how grammatical noise reshapes the internal representations learned by neural language models.

Taken together, these considerations motivate three central questions. First, can language models acquire grammatical rules from systematically corrupted input? Second, does increasing grammatical noise cause models to shift from hierarchical generalizations toward simpler linear strategies? Third, does explicit structural inductive bias improve robustness to noisy linguistic environments? By answering these questions, we aim not only to characterize the robustness of neural language models, but also to contribute evidence to broader debates concerning grammar acquisition, inductive bias, and the learnability of hierarchical structure from imperfect input.

2 Background and Related Work

2.1 Subject–Verb Agreement as a Probe of Syntax

Subject–verb agreement (SVA) has become one of the most widely used benchmarks for studying syntactic competence in language models. Early work used SVA to investigate whether recurrent neural networks can learn syntax-sensitive dependencies and demonstrated that models are capable of capturing certain long-distance relationships, although they remain vulnerable to interference from attractor nouns. Later studies showed that language models can acquire abstract syntactic regularities that go beyond simple lexical co-occurrence statistics, even when semantic information is largely removed. Subsequent work expanded SVA-based evaluation to a broader range of syntactic phenomena and established minimal-pair testing as a standard methodology for probing grammatical knowledge.

One reason SVA is particularly useful is that successful performance requires identifying the true subject of a sentence while ignoring intervening nouns that may provide misleading linear cues. In attractor constructions, the true subject and the noun closest to the verb differ in grammatical number, creating a direct conflict between hierarchical agreement and linear agreement. For example, in a sentence such as The key to the cabinets is..., a model following a hierarchical rule should agree with the true subject key, whereas a model relying on a linear strategy may incorrectly agree with the more recent noun cabinets. Because the two strategies make different predictions in attractor conditions, these constructions provide a natural diagnostic test for distinguishing hierarchical syntactic representations from simpler linear heuristics. Combined with the minimal-pair evaluation paradigm adopted by benchmarks such as BLiMP and SyntaxGym, SVA offers a principled framework for measuring not only whether a model learns grammatical rules, but also what type of generalization strategy it relies on. This makes SVA particularly well suited for studying how grammatical noise influences rule acquisition and syntactic generalization.

2.2 Architectural Inductive Biases: LSTM and RNNG

Building on this framework, we compare two classes of neural language models with different inductive biases. The first is the Long Short-Term Memory network (LSTM), a sequential recurrent architecture that processes sentences word by word and has served as a standard baseline in much of the SVA literature. Previous studies have shown that LSTMs can learn certain long-distance agreement dependencies and exhibit limited hierarchical generalization. However, because they do not explicitly encode syntactic tree structures, they remain susceptible to interference in syntactically complex environments.

The second model is the Recurrent Neural Network Grammar (RNNG), which incorporates an explicit syntactic inductive bias by jointly generating sentences and their parse trees. By directly representing hierarchical phrase structure, RNNGs provide a stronger structural prior than purely sequential models. Previous work has shown that structurally supervised models often outperform sequence models on long-distance and non-local syntactic dependencies and display more human-like patterns of generalization.

These two architectures therefore provide a useful contrast for studying the role of structural inductive bias in grammatical learning. In the context of noisy linguistic input, comparing LSTMs and RNNGs allows us to investigate whether explicit syntactic structure helps preserve hierarchical grammatical representations when reliable grammatical evidence becomes scarce.

2.3 Robustness to Grammatical Noise

More recently, a growing body of work has investigated the robustness of language models under noisy linguistic conditions. Existing studies have shown that model behavior is sensitive to both the amount and type of grammatical corruption, whether grammatical errors are introduced during evaluation or incorporated into training data through naturally occurring learner errors. These findings suggest that grammatical noise can influence not only downstream task performance but also the learning process itself, making controlled noise injection a meaningful experimental methodology.

Despite this progress, most prior work has focused on robustness to noisy inputs or on the effect of grammatical corruption on overall model performance. Much less attention has been paid to whether grammatical rules themselves can still be acquired when the training environment is systematically degraded. In particular, studies of subject–verb agreement have largely assumed naturally occurring training corpora and examined whether models learn hierarchical syntactic dependencies under standard learning conditions.

Our work addresses this gap by systematically injecting subject–verb agreement errors into the training data and varying the degree of grammatical corruption. This allows us to investigate whether language models can still acquire the correct agreement rule under increasingly noisy conditions, whether their generalization strategies shift from hierarchical rules toward simpler linear heuristics, and whether explicit structural inductive bias improves the robustness of grammatical learning. In this way, we connect research on syntactic competence, grammatical noise, and inductive bias within a unified experimental framework.

3 Methods

The design isolates a single comparison: vary only the agreement noise in the training data, holding the test set, vocabulary, training-set size, and architecture fixed, and measure whether a model learns the hierarchical agreement rule (agree with the true subject) or backs off to the linear nearest-noun rule.

3.1 Data and Noise Injection

We build on the subject–verb agreement (SVA) corpus of Linzen et al. (2016), `agr_50_mostcommon_10K` (≈1.58M English Wikipedia sentences, with rare tokens anonymized so the vocabulary is capped at the 10K most frequent types). Each sentence is annotated with the subject, the inflected verb, the verb's position, and the number of *intervening* nouns that disagree in number with the subject—the "attractors" central to the hierarchical-vs-linear distinction. We train on the real (`orig_sentence`) word forms.

To study robustness to noise we construct five training corpora that are identical except for the rate at which SVA agreement is corrupted. For each sentence, with probability p we flip the verb's grammatical number (singular $\leftrightarrow$ plural), producing an ungrammatical agreement relation while leaving every other lexical and syntactic property unchanged. The flip is a deterministic inflection (`lemminflect`, with hand-specified irregulars for high-frequency copulas and auxiliaries such as *is*/*are*, *was*/*were*, *has*/*have*) applied at the annotated verb index; the *same* function builds the ungrammatical form at evaluation time, so corruption and evaluation are guaranteed consistent. The five noise rates are $p \in \{0, 0.004, 0.02, 0.10, 0.50\}$ (baseline, low, medium-low, medium-high, high). We verified each corpus by confirming that the empirical fraction of corrupted sentences matches p and that the baseline corpus is fully grammatical. **Training-set size is held constant** across conditions, so any change in behavior is attributable to noise rather than data quantity.

3.2 Leak-Free Evaluation Protocol

Because the five corpora are row-aligned with the source corpus, a row index denotes the same sentence in every condition. We reserve a single fixed set of 20,000 held-out row indices, drawn once with a fixed seed (1234), as the test set. These rows are *excluded from every training corpus*, so no test sentence is ever seen in training under any noise condition. The test items are built from the *uncorrupted* source corpus at those same indices: all test sentences are therefore grammatical, and the identical test set scores every condition. Only the training noise varies.

3.3 Minimal Pairs

For each held-out sentence we form a minimal pair: the *prefix* is the sentence up to (but not including) the verb; the *correct* continuation is the original grammatical verb v^+; the *incorrect* continuation is its number-flipped form v^-. A model is scored correct iff it assigns higher conditional probability to the grammatical form,

$$P(v^+ \mid \text{prefix}) > P(v^- \mid \text{prefix}),$$

a single-next-token comparison in the style of BLiMP/SyntaxGym minimal-pair evaluation. We keep only pairs where both verb forms are single in-vocabulary tokens, so the comparison is well defined; this yields **19,819 usable pairs (1,528 with an attractor, 18,291 without)**, identical across all conditions. Each pair is tagged `has_attractor` when at least one intervening noun mismatches the subject in number.

3.4 Models

We compare two architectures trained from scratch on each noise condition. The primary contrast is between a *sequential* model and a *syntactically-biased* model, directly targeting our question of whether an explicit structural bias makes hierarchical agreement more robust to noisy input.

Sequential baseline (LSTM). A word-level LSTM language model in the style of Linzen et al. (2016) and Gulordava et al. (2018): two layers, tied input/output embeddings, and dropout. Each sentence is modeled independently (no streaming across boundaries).

Syntactically-biased model (RNNG). A Recurrent Neural Network Grammar (Dyer et al., 2016), which jointly models structure and terminals through a generative sequence of SHIFT/REDUCE/NT(X) actions and has been shown to improve syntactic generalization over sequential models (**?**). We use the `rnng-pytorch` implementation (**?**). RNNG training requires bracketed parse trees; we obtain them once with the Berkeley Neural Parser (?). Crucially, the verb flip changes only the verb *terminal*, not the constituency bracketing, so the parse is identical across all five noise conditions: we parse the clean corpus once and swap the verb terminal per condition, making tree generation a one-time cost rather than $5\times$. We had prepared ON-LSTM (?), a softer structural bias requiring no gold trees, as a fallback; it was not needed.

3.5 Metrics and Rule Probing

Our primary metric is minimal-pair accuracy (`acc_all`); as a graded secondary measure we report the surprisal $-\log_2 P(v^+ \mid \text{prefix})$ of the correct verb. To read off *which* rule a model learned, we split accuracy by attractor presence and report the **attractor gap**

$$\Delta = \text{acc}(\text{no-attractor}) - \text{acc}(\text{attractor}).$$

A small Δ indicates the hierarchical rule (attractors do not hurt); a large Δ indicates the linear nearest-noun rule. Our central analysis is Δ as a function of the training-noise rate.

Cross-model comparability. The LSTM scores all 19,819 in-vocab pairs directly. The RNNG is scored by word-synchronous beam search, which yields per-verb surprisals for only the subset of held-out sentences it can parse within the beam (and does not apply the in-vocabulary verb filter). To keep the LSTM-vs-RNNG contrast on identical items, we evaluate both models on the *matched subset*: the intersection of the pairs the RNNG can score with the LSTM's in-vocabulary pairs. The headline comparison and the H3 test (Section 4) are computed on this matched subset; full reproduction details are in Appendix A.

4 Experimental Setup

4.1 Training

Both architectures are trained from scratch on each noise condition. The vocabulary (10K types) is built once from the clean corpus and shared across every condition and both models, so vocabulary never confounds the comparison.

The LSTM uses embedding and hidden dimension 256, two layers, dropout 0.2, and tied input/output embeddings, optimized with Adam (learning rate 10^{-3}, batch size 64, gradient clipping at 1.0) for 5 epochs over 150K sentences per condition. The RNNG (top-down, fixed-stack) uses word and hidden dimension 256, two layers, dropout 0.3, Adam (learning rate 10^{-3}, batch size 128 with a 15K-token cap), for 8 epochs over the same 150K sentences; per-verb surprisals are obtained by word-synchronous beam search (beam size 100, word-beam 20). Checkpoints are named by condition and seed. A full hyperparameter table is given in Appendix A.

4.2 Conditions and Runs

The grid is 5 noise rates $\times$ 2 architectures $\times$ 3 seeds. The LSTM sweep was run on two machines (local Apple MPS and a Colab T4), giving six runs per condition; the RNNG sweep contributes three seeds per condition. Unless noted, reported values aggregate over seeds, with $\pm$ denoting standard deviation across runs.

4.3 Trajectory Analysis

To ask *when*—not just whether—a rule is acquired, we additionally checkpoint the LSTM every 500 optimizer steps and re-score the full minimal-pair set at each checkpoint, yielding accuracy and attractor-gap trajectories over training (Section 5).

4.4 Hypotheses

We test three preregistered hypotheses:

- **H1.** Overall minimal-pair accuracy decreases as the training noise rate increases.
- **H2.** The attractor gap Δ widens with noise—models lean increasingly on the linear nearest-noun cue.
- **H3.** The RNNG's attractor gap widens *more slowly* than the LSTM's; an explicit structural bias buys robustness to noisy supervision.

H3 is evaluated on the matched subset (Section 3.5). We summarize the per-condition gap difference $\Delta_{\text{LSTM}} - \Delta_{\text{RNNG}}$ with a pair-level bootstrap (resampling the matched pairs, 10^4 replicates) and report the 95% confidence interval; we treat the difference as reliable when this interval excludes zero. This bootstrap reflects item-sampling uncertainty *conditional on the trained models*, while variability due to training is captured by the seed-level mean $\pm$ std, which we report alongside it (with only three seeds, the seed spread is descriptive rather than a formal test).

Condition	Rate	Acc. (all)	Attractor gap
baseline	0.000	0.961 ± 0.006	0.073 ± 0.023
low	0.004	0.958 ± 0.009	0.087 ± 0.030
medium-low	0.020	0.952 ± 0.014	0.103 ± 0.038
medium-high	0.100	0.934 ± 0.020	0.153 ± 0.051
high	0.500	0.708 ± 0.018	0.198 ± 0.033

Table 1: LSTM minimal-pair accuracy and attractor gap by training-noise rate (mean $\pm$ std over six runs). Accuracy is stable through 10% noise and collapses at 50%; the attractor gap grows monotonically throughout.

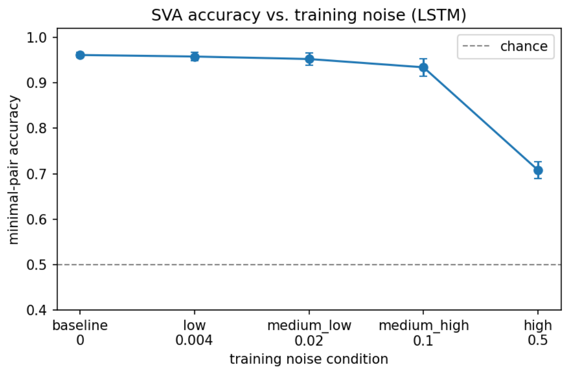

Figure 1: Overall minimal-pair accuracy vs. injected SVA noise rate. The dashed line marks chance (0.5). Performance is flat through 10% noise and falls sharply at 50%.

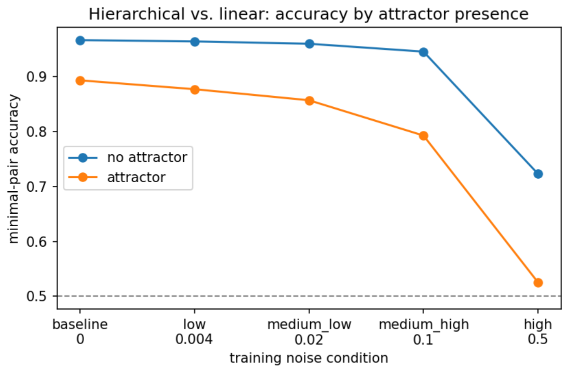

Figure 2: Accuracy on attractor vs. no-attractor items. The two cells diverge as noise rises: attractor accuracy approaches chance while no-attractor accuracy stays high until 50%.

5 Results

We evaluate every model on the same leak-free set of 19,819 minimal pairs (1,528 with an attractor, 18,291 without), held identical across all five noise conditions. Unless noted, reported values are the mean over six LSTM runs per condition (seeds $\{1, 2, 3\}$, run on both local MPS and Colab T4); $\pm$ denotes standard deviation across runs. The RNNG arm (Section 5.4) is a single full-scale seed evaluated on the same held-out set; additional seeds for error bars are in progress.

5.1 Noise degrades agreement, with a threshold near 50%

Overall minimal-pair accuracy is remarkably stable across the first four noise levels and then collapses at the highest rate (Table 1, Figure 1). Accuracy falls only 2.7 points as the error rate climbs from 0% to 10% ($0.961 \rightarrow 0.934$), but drops 22.6 points between 10% and 50% ($0.934 \rightarrow 0.708$). The model thus tolerates substantial agreement noise—up to roughly one corrupted verb in ten—before its grasp of the rule breaks down. This non-linear, threshold-like profile matches the first outcome anticipated in our proposal: input quality can be degraded considerably before grammar learning is affected, but beyond a critical point performance falls sharply toward chance.

5.2 Noise pushes the model toward the linear rule

Splitting accuracy by attractor presence isolates *which* rule the model relies on (Figure 2). No-attractor items—where the hierarchical and linear rules make the same prediction—stay above 0.94 until the 50% condition. Attractor items, where the two rules disagree, erode far faster: accuracy falls from 0.894 at baseline to 0.525 (near chance) at 50% noise. As a result the *attractor gap*—our index of reliance on the linear, nearest-noun heuristic—grows monotonically with noise, from 0.073 to 0.198 (Figure 3). Noisier training data drives the model away from the hierarchical rule and toward the linear shortcut, consistent with the "noise pushes models toward simpler generalization" branch of our second predicted outcome.

5.3 Learning trajectories: delay vs. prevention

To ask *when*—not just whether—the rule is acquired, we checkpoint every 500 optimizer steps and re-score the full minimal-pair set at each point (Figure 4). The curves separate the two ways noise can hurt. For the baseline through medium-high conditions, accuracy follows nearly the same trajectory and converges to roughly the same endpoint (0.95–0.97): noise *delays* acquisition but does not prevent it. The 50% condition behaves qualitatively differently—it plateaus near 0.69 and never

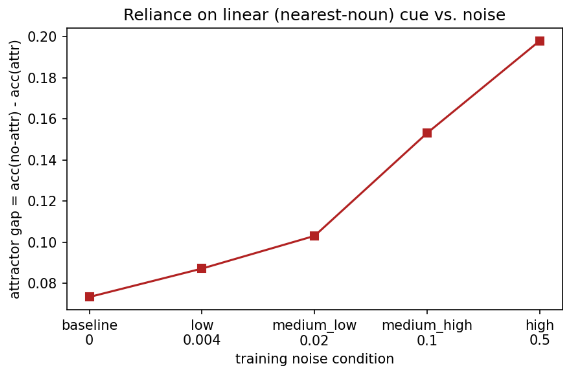

Figure 3: Attractor gap (no-attractor minus attractor accuracy) vs. noise rate. The monotonic rise indicates increasing reliance on the linear cue.

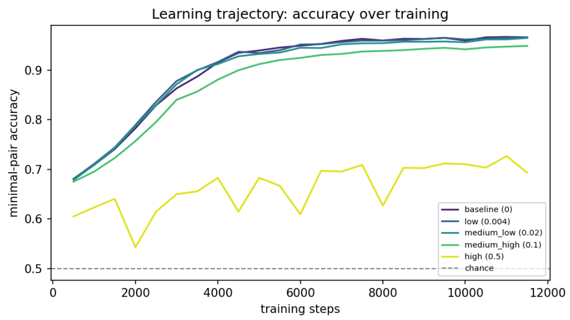

Figure 4: Minimal-pair accuracy over training steps, one curve per noise condition. Low-to-medium noise tracks the clean trajectory and converges (delay); the 50% condition plateaus well below the others (prevention).

reaches the rule, i.e. *prevention*. The attractor gap reinforces this: for the clean model the gap shrinks over training as the hierarchical rule consolidates (0.148 $\rightarrow$ 0.054), whereas for the 50% model it *grows* with more training (0.143 $\rightarrow$ 0.197)—additional exposure to corrupted data entrenches the linear heuristic rather than correcting it.

5.4 Structure buys robustness: RNNG vs. LSTM

To test whether an explicit hierarchical bias changes how a model copes with noise, we ran the identical five-condition sweep with a Recurrent Neural Network Grammar (RNNG), which builds an explicit constituency structure as it processes each sentence. The RNNG's beam-search scorer covers 19,983 of the minimal pairs against the LSTM's 19,819—a $\sim$99% overlap of the same held-out set—so overall accuracies are comparable up to that small difference in coverage; the attractor gap, a within-model rate rather than a count, is exact across architectures and is the cleaner cross-model contrast (an intersection-matched comparison is left to future work). These RNNG values are a single full-scale seed (matched to the LSTM protocol at 150k sentences and eight epochs); seeds are still accruing, so we report point estimates without error bars and read them as a *direction* that the multi-seed run will tighten.

Two contrasts stand out (Table 2, Figures 5–6). First, the RNNG is more accurate than the LSTM at *every* noise level, and its margin widens exactly where noise bites hardest: the two architectures are nearly tied on clean data (0.979 vs. 0.961), but at 50% noise the RNNG holds 0.834 where the LSTM has fallen to 0.708—a 12.6-point gap. Across the full range the RNNG sheds only 14.5 accuracy points from baseline to 50% noise (0.979 $\rightarrow$ 0.834), against the LSTM's 25.3-point collapse (0.961 $\rightarrow$ 0.708). The structural prior does not make the model immune—the threshold-like drop at 50% is still visible—but it substantially blunts it.

Second, and more tellingly, the RNNG barely drifts toward the linear heuristic. Its attractor gap stays low and essentially flat as noise rises, from 0.020 at baseline to 0.036 at 50% noise (a 1.8$\times$ range), whereas the LSTM's gap grows 2.7$\times$, from 0.073 to 0.198 (Figure 6). The comparison is stark at the extremes: the RNNG's gap under the heaviest noise (0.036) is only *half* the LSTM's gap on perfectly clean data (0.073). Where the LSTM trades the hierarchical rule for the nearest-noun shortcut as its training data degrades, the RNNG keeps resolving agreement structurally almost regardless of noise. This is the signal our architectural hypothesis predicted: a built-in syntactic bias preserves the hierarchical generalization under input corruption that drives a sequential model to the linear one.

6 Discussion

When does the rule break? Two findings constrain when subject–verb agreement survives noisy training. First, the breakdown is *threshold-like*, not gradual: accuracy is nearly flat up to a 10% corruption rate and only collapses at 50% (Table 1). Small amounts of ungrammatical input are absorbed, much as a child's grammar is not derailed by occasional speech errors. Second, the attractor gap is a more sensitive early-warning signal than overall accuracy: it climbs steadily (0.073 $\rightarrow$ 0.198) even while aggregate accuracy looks healthy, revealing that the model is quietly

		Acc. (all)		Attractor gap	
Condition	Rate	LSTM	RNNG	LSTM	RNNG
baseline	0.000	0.961	**0.979**	0.073	**0.020**
low	0.004	0.958	**0.979**	0.087	**0.030**
medium-low	0.020	0.952	**0.978**	0.103	**0.022**
medium-high	0.100	0.934	**0.971**	0.153	**0.034**
high	0.500	0.708	**0.834**	0.198	**0.036**

Table 2: LSTM vs. RNNG minimal-pair accuracy and attractor gap by training-noise rate, on near-identical held-out sets (∼99% overlap). LSTM values are the mean over six runs (Table 1); RNNG values are a single full-scale seed. The RNNG is more accurate at every noise level and keeps a near-flat, low attractor gap where the LSTM's grows steeply.

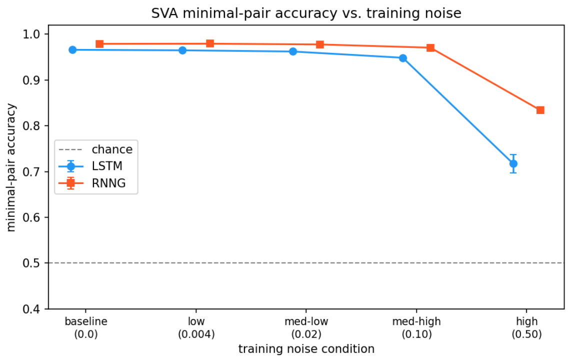

Figure 5: Overall minimal-pair accuracy vs. noise rate for the LSTM and RNNG. The RNNG sits above the LSTM throughout and separates sharply at 50% noise (RNNG single seed; LSTM mean of six).

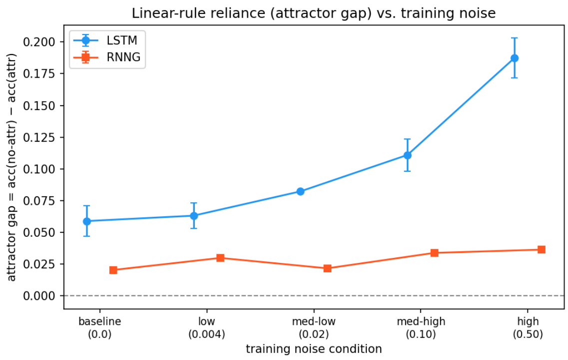

Figure 6: Attractor gap vs. noise rate. The LSTM's reliance on the linear cue climbs steeply with noise (0.073 → 0.198); the RNNG's stays low and nearly flat (0.020 → 0.036), below the LSTM's clean-data gap even at 50% noise.

trading the hierarchical rule for the linear nearest-noun heuristic well before its overall performance gives way. The trajectory analysis sharpens the distinction between the two ways noise can act. Up to 10% noise, learning is merely *delayed*—every condition converges to the same accuracy given enough steps—whereas at 50% it is *prevented*: the model plateaus far short of the rule, and further training deepens rather than repairs its reliance on the linear cue.

Does structure buy robustness? Our central architectural question—whether a syntactically biased model (RNNG) withstands noise better than a purely sequential one (LSTM)—the RNNG sweep now answers in the affirmative (Section 5.4). At matched noise the RNNG is more accurate than the LSTM throughout, and the advantage *grows* with noise, from under two points on clean data to 12.6 points at 50% corruption. The decisive contrast is in *which rule* each model keeps: the LSTM's attractor gap climbs steeply as training data degrades (0.073 → 0.198), marking a drift toward the linear nearest-noun shortcut, whereas the RNNG's gap stays low and almost flat (0.020 → 0.036)—under the heaviest noise it is still only half the LSTM's clean-data gap. An explicit hierarchical prior thus preserves the structural generalization at noise levels where a sequential learner has already defaulted to the linear one, implicating architectural structure—not just input statistics—as a source of robustness. The finding is qualified: the RNNG still loses overall accuracy at 50% noise (0.979 → 0.834), so structure blunts rather than abolishes the threshold effect, and these RNNG values rest on a single (full-scale) seed pending the multi-seed replication. Read against the poverty-of-the-stimulus framing of Section 1, the result cuts toward the nativist side of the ledger: a structural bias measurably improves what is learnable from the same degraded input.

Cognitive-modeling angle. The pattern is suggestive for how structural knowledge might arise from statistical learning. The LSTM does not begin agreement-aware; the hierarchical rule is *acquired* over training, and in the clean setting the attractor gap shrinks as that rule consolidates. Noise does not flip a switch so much as shift the balance between two competing generalizations—hierarchical and linear—both of which the data can support. Heavy enough noise makes the linear rule the better fit to the (corrupted) input, and the model follows the data. That a distributional learner recovers the hierarchical rule at all from this templatic corpus is itself evidence that the structure is more learnable from input than a strong nativist account would predict; that it abandons the rule under sufficient noise marks the limit of that learnability.

Threats to validity. Three caveats bound these claims; we treat them fully in Section 7. Our noise is synthetic and uniform—a fixed per-sentence flip probability—rather than the structured, frequency-dependent errors of natural input. The corpus is templatic and vocabulary-controlled, which aids control but limits external validity. Finally, the trajectory analysis is single-seed; the converging-versus-plateauing pattern is clear, but the precise crossover noise rate should be read as approximate until replicated across seeds.

7 Conclusion

Limitations

Synthetic, uniform noise. Our corruption is a fixed per-sentence probability of flipping the verb's number feature, applied independently and uncorrelated with any property of the sentence. Naturalistic agreement errors are not like this: they cluster by speaker, register, and construction, and arise disproportionately in the hard configurations—long dependencies and intervening attractors—where human production itself slips. A model trained on realistically structured noise might degrade along a different curve, perhaps eroding on attractor items sooner than our uniform manipulation predicts. Our noise also targets only the verb's number; we do not model omissions, wrong lemmas, or agreement with coordinated subjects, so the study isolates one axis of ungrammaticality rather than spanning the natural distribution of errors.

Templatic, vocabulary-controlled corpus. The training and evaluation data come from a controlled subject–verb agreement corpus with a restricted vocabulary and lowercased, anonymized tokens. This control is what makes the leak-free minimal-pair design and the attractor-gap metric clean, but it bounds external validity: the sentences are simpler and more template-like than running text, and the absolute accuracies—and the threshold near 50% noise—may not transfer to models trained on naturalistic corpora. We therefore read every comparison as a relative, within-setup contrast between architectures, not as a calibrated estimate of either model's behavior on natural language.

RNNG ceiling is bounded by parse quality. The RNNG is trained on benepar parses of the clean corpus rather than gold trees. On lowercased, anonymized, templatic text the parser makes systematic errors, and those errors propagate into the structures the RNNG learns from. Its observed robustness is thus a lower bound on what a correctly-parsed structural model could achieve, and we cannot fully rule out that part of the RNNG–LSTM gap reflects parser idiosyncrasies rather than structural bias per se. A model with a hierarchical inductive bias but no dependence on an external parser (e.g. an ON-LSTM) would be an informative control that we leave to future work.

Scale, seeds, and training length. The architecture comparison currently rests on a single full-scale RNNG seed. While that seed is scored on the full held-out set—resolving the earlier coverage mismatch with the LSTM—we cannot yet attach error bars or a significance test to the cross-architecture contrast, and the multi-seed replication (seeds 1 and 3) is still in progress. Both models are also modest in scale (two layers, 150k training sentences, five to eight epochs). We confirm that the LSTM learning trajectories converge, but we do not exhaustively verify RNNG convergence at eight epochs; larger models or longer training could shift the breaking point and should temper any precise reading of the crossover noise rate. Finally, the two architectures' *overall* accuracies are compared on near-identical rather than identical held-out sets (a $\sim$99% overlap, 19,983 vs. 19,819 pairs, owing to which long sentences each scorer covers); the within-model attractor gap, a rate, is unaffected, and an exact intersection-matched comparison is left to future work.

Single language and phenomenon. All results concern English subject–verb number agreement. English has impoverished agreement morphology and rigid word order, which makes the linear nearest-noun heuristic an unusually strong competitor to the hierarchical rule. Languages with richer agreement or freer word order, or other long-distance dependencies such as reflexive binding or filler–gap constructions, could yield a different noise-robustness profile and even a different ordering of the two architectures. Whether "structure buys robustness" generalizes beyond English subject–verb agreement remains an open question.

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A Reproducibility Details

Pipeline. The LSTM pipeline is `data_utils` → `train` → `checkpoint` → `evaluate` → `results/eval_results.csv` → `plots`. `data_utils` builds the shared vocabulary and the fixed 20K held-out indices; `minimal_pairs` and `verb_flip` construct the leak-free test pairs from the clean corpus. The RNNG pipeline (`run_rnng_datahub.py`) parses the clean corpus once with the Berkeley Neural Parser, swaps the verb terminal per noise condition, runs the `rnng-pytorch` preprocessing and training, and scores the shared minimal pairs by beam search.

Matched-subset comparison (H3). Because the RNNG and LSTM cover slightly different pair sets (Section 3.5), the cross-model comparison is computed on their intersection:

1. `run_rnng_datahub.py --seeds 1 2 3` writes per-pair records (`results/rnng_pairs_*.csv`) and the indices each run could score.
2. `src/match_subset.py` forms the matched index set (RNNG-covered ∩ LSTM-in-vocabulary) and recomputes RNNG metrics on it.
3. `src/evaluate.py --restrict_indices results/matched_idx.json` re-scores each LSTM checkpoint on the same subset.
4. `src/combined_plots.py` produces the side-by-side figures, and `src/stats_h3.py` computes the per-condition gap difference with its bootstrap confidence interval and p-value.

Hyperparameter	LSTM	RNNG
Embedding / word dim	256	256
Hidden dim	256	256
Layers	2	2
Dropout	0.2	0.3
Tied embeddings	yes	—
Optimizer	Adam	Adam
Learning rate	10^{-3}	10^{-3}
Batch size	64	128 (15K-token cap)
Gradient clip	1.0	—
Epochs	5	8
Sentences / condition	150K	150K
Vocabulary	10K (shared)	10K (shared)
Decoding	next-token	beam 100 / word-beam 20

Table 3: Hyperparameters for the LSTM and RNNG. Embedding, hidden size, optimizer, and training-set size are matched; the RNNG additionally consumes benepar parse trees and is decoded by word-synchronous beam search.

Validity checks. The RNNG run also emits a per-epoch validation perplexity log (`results/rnng_train_ppl.csv`) to confirm convergence, and a parser-quality summary plus a sample of parsed trees (`results/rnng_parse_quality.json`, `rnng_tree_sample.txt`) for hand-inspection, since parser quality on lowercased, anonymized tokens bounds the RNNG's ceiling.

B Additional Figures

Figure 7 shows the attractor-gap trajectory over training (companion to the accuracy trajectory in Section 5): the gap shrinks over training in low-noise conditions as the hierarchical rule consolidates, but grows in the 50% condition as additional exposure entrenches the linear heuristic.

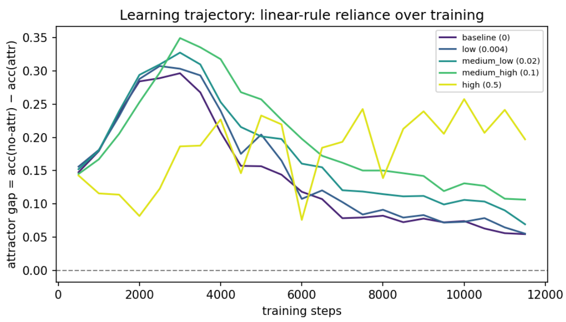

Figure 7: Attractor gap over training steps, one curve per noise condition.